Introduction to Retrieval-Augmented Generation

A short test document for RAGdocAI

What is RAG?

Retrieval-Augmented Generation, commonly called RAG, is an approach that combines information retrieval with text generation. A system first retrieves relevant information from a knowledge source and then provides that information to a language model.

Document Processing

A document-based RAG system may extract text from PDF files, split the text into smaller chunks, convert chunks into embeddings, and store those embeddings in a vector database.

Embeddings and Vector Search

An embedding is a numerical representation of text. Vector search compares embeddings to find text passages that are semantically similar to a user's question.

Answer Generation

After retrieving relevant passages, the system sends the question and retrieved context to a language model. The model uses that context to produce an answer. A well-designed RAG system should avoid inventing information that is not supported by its documents.

Prepared as sample content for local RAGdocAI testing.